Natural selection

BACKGROUND

genes (say 'jeans') – information carried in a living thing's cells that determines some of its characteristics

Natural selection is the term we give to the way in which a particular animal or plant survives to breed. Natural selection results in species **adaptation**. Evolutionary scientists call this process **evolution by natural selection**.

Some animals within a species – a group of rabbits, for example – will be healthier, stronger and quicker than others. If this group of rabbits were to be chased by a wolf or a fox, the 'fittest' ones would survive to breed because they could run faster and for longer periods of time. These better-adapted rabbits would eventually have babies, and most of the young rabbits might also inherit their parents' strength and fitness. Over time, the less fit rabbits would die out and the stronger, fitter, faster rabbits would survive. Nature is 'selecting' those rabbits to be the ones that pass their genes down to a new generation.

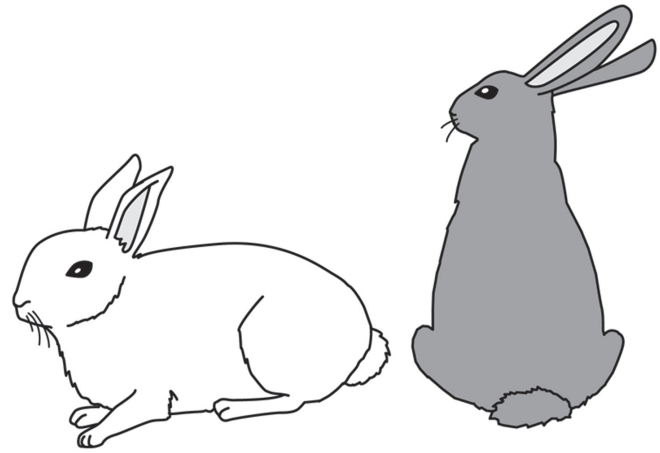
☞ **1** Complete this sentence, using your own words.

The word 'fittest' means _____.

2 Use just one word from the box to fill the spaces, and complete the passage.

diseases	babies	parents	rabbits

The _____ of the 'fittest' creatures will be more likely to survive to have _____ of their own. It is highly likely that those _____ will inherit the 'stronger' features of their parents. They will also probably pass those genes on to their own _____. There will be less and less individuals with 'weaker' features, as those with 'stronger' features dominate. The 'weaker' individuals will die out and only the 'fittest' will remain.



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Natural selection CONT'D

3 Which of these rabbits show features that would most likely be 'selected' and help a group of rabbits survive? Circle your choices.

a) this small, weak male rabbit loses contests with stronger male rabbits to mate with a female

b) this rabbit with white fur is able to hide better in its snowy habitat than a rabbit with dark fur

c) this pet rabbit is chased by a dog and is killed

d) this rabbit is able to run faster than another when it is being chased by a wolf

e) this rabbit becomes blind

f) this rabbit is healthier than others and is able to survive through a drought

g) this rabbit fails to build itself a burrow before a snowy winter sets in

h) this rabbit population has larger teeth that allow the rabbits to chew thicker, more nutritious grass than other smaller-toothed rabbits

i) this male rabbit has shorter fur that allows it to stay cooler in its hot desert habitat than furrier rabbits

j) this group of rabbits has chosen to live in an area that is a popular hunting ground where people come to shoot rabbits

k) this rabbit's family built a better network of burrows than another group

l) this pet rabbit refuses to eat vegetables or plants of any kind

m) this rabbit has its burrows in an area where foxes begin to roam

n) this rabbit has immunity to a disease that kills other rabbits

o) this rabbit has fur patterns that make it less attractive to mating partners than other rabbits

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